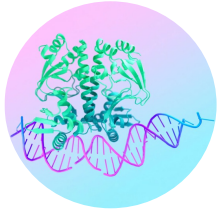


AI agents in action



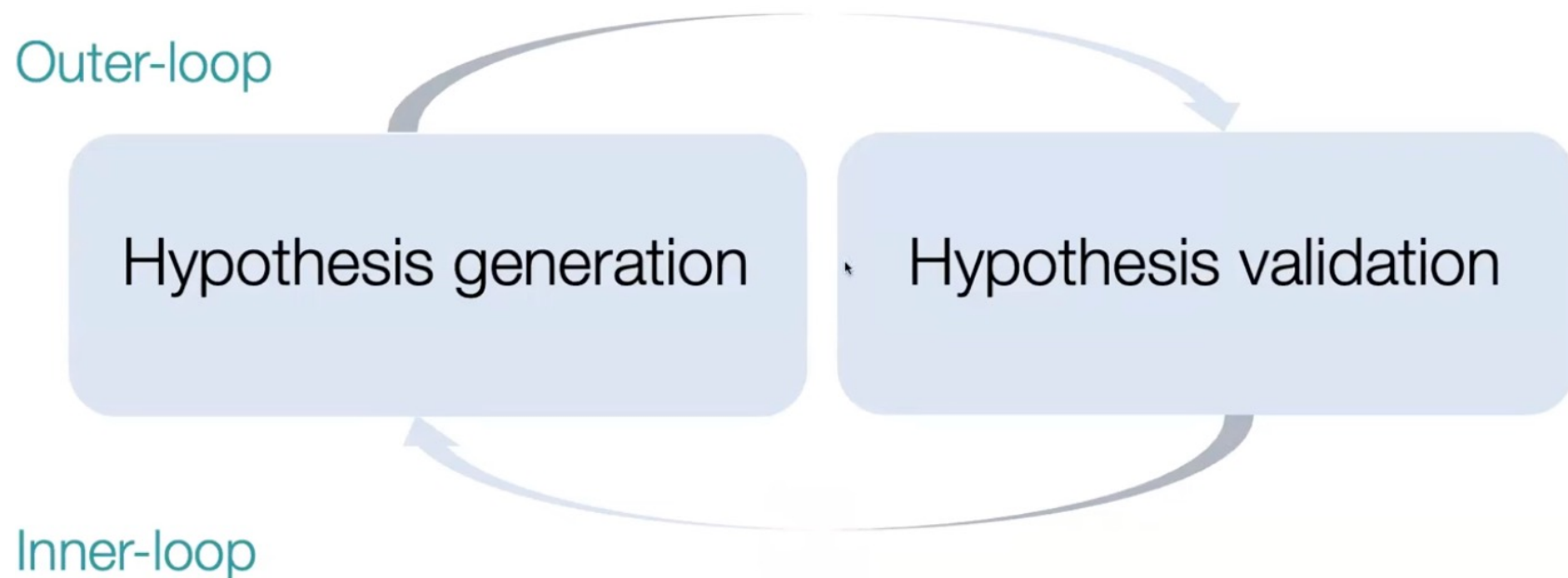
Mahbub Ul Alam
Data Engineer (AI)
SciLifeLab Data Centre
mahbub.ul.alam@scilifelab.se



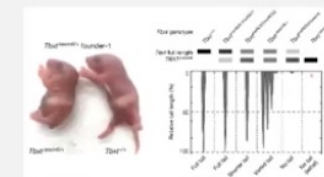
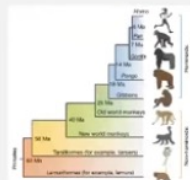
Life science research is a loop of reasoning and actions



Every discovery cycles between thinking and doing.



Massive number of individual workflows



What we want from a life science AI agent



Six things that turn a chatbot into a research collaborator.

Capability	Details
Adapt	Plug in tools from genomics, drug discovery, clinical data, ... any domain
Unify	Orchestrate workflows across multiple domains
Evolve	Add new skills as research advances
Efficient	Automate repetitive data tasks
Reproducible	Transparent, executable record of every step
Scalable	Tackle complex, multi-step research at scale

Two ways to put an agent to work



Pick by how well an existing agent fits your problem.



Path 1, Adopt: your task matches a ready-made agent? Use it. Fastest route, nothing to build.

Path 2, Build: need your own data, tools, or workflow? Assemble your own.

The dividing line is the tools: ready-made agents **bring their own**, a **custom agent wraps yours** (your scripts, databases, APIs, even a whole deployed app).

Path 1, find the right agent for the job

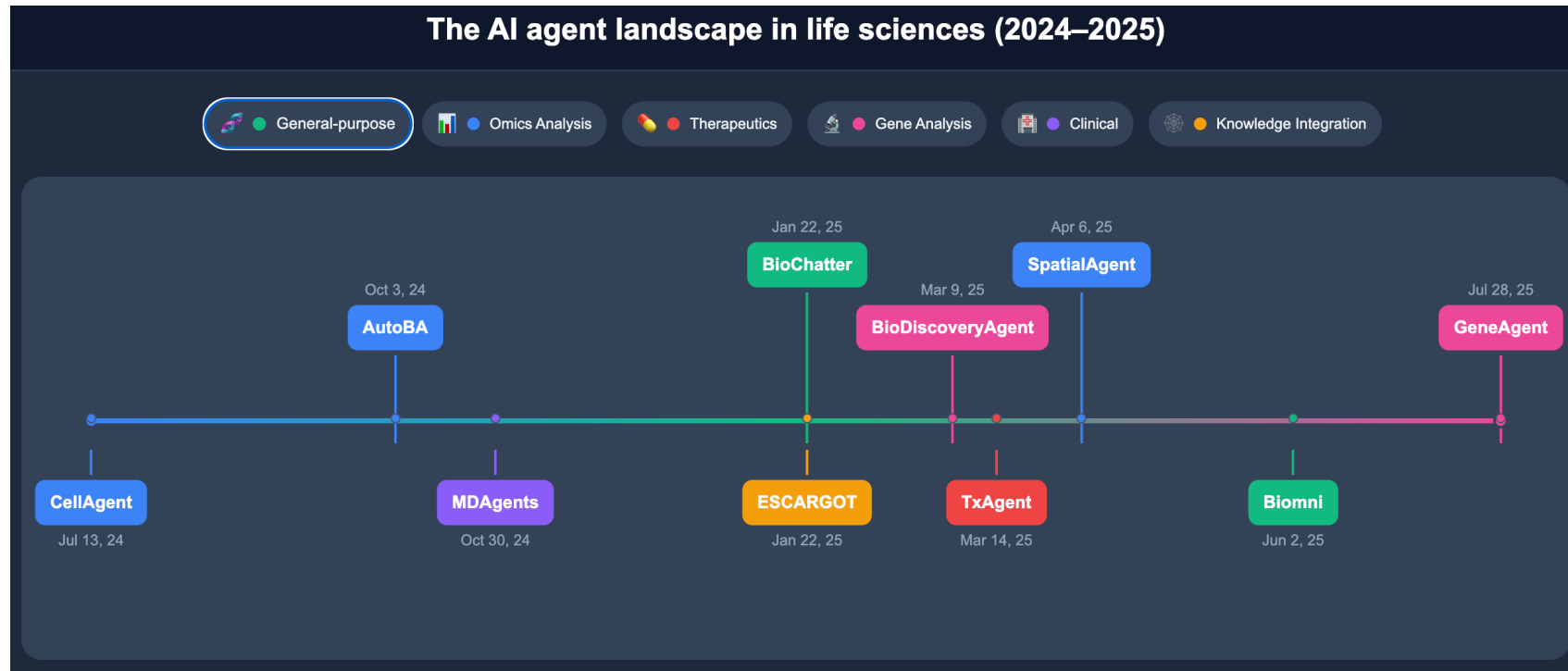


Chances are, someone already built one for your task.



Tool 1

shorturl.at/MOY7d

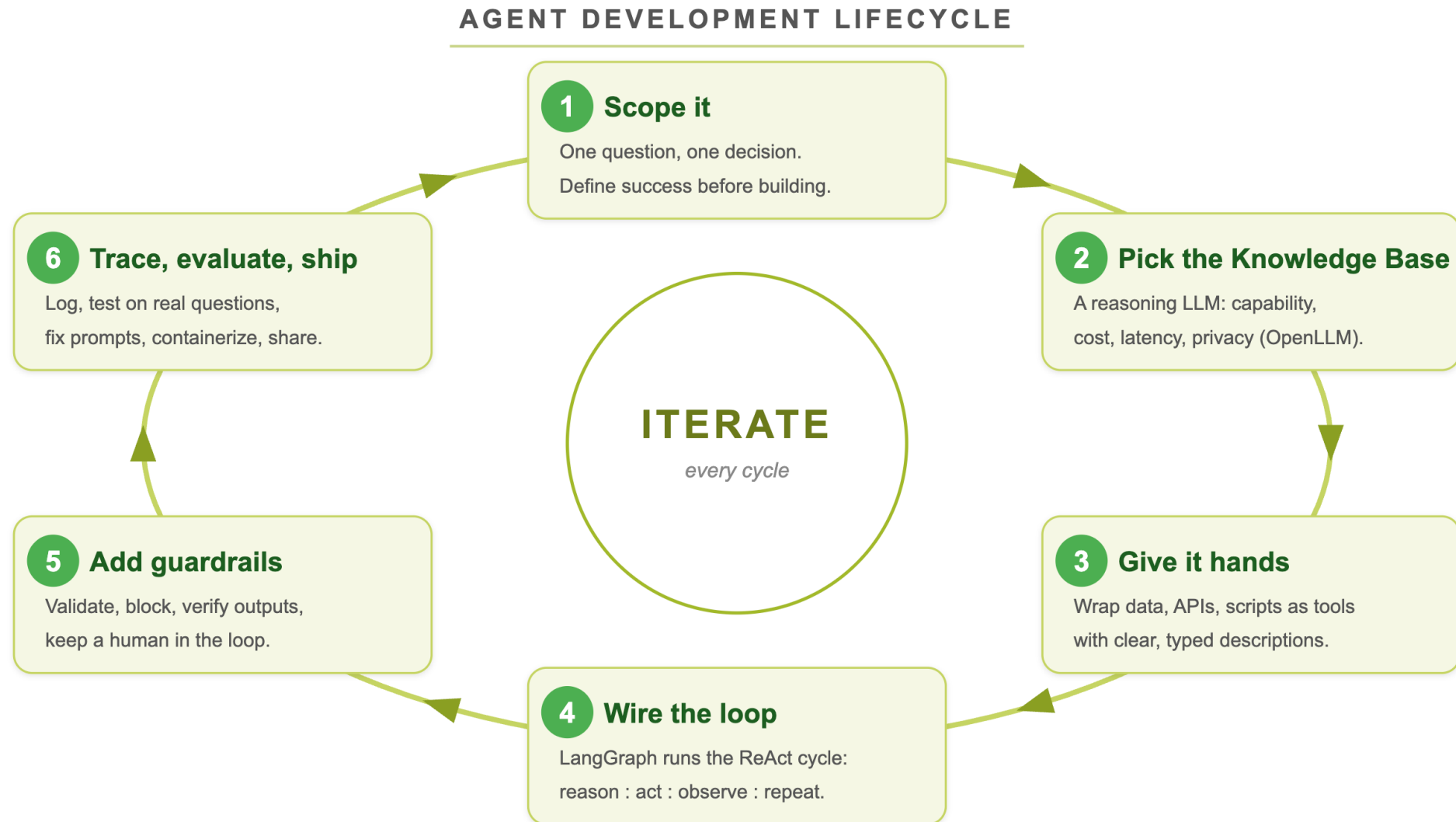


Tool 2

shorturl.at/kxydE

Explore two AI agent landscape tools

Path 2, a typical agent development workflow



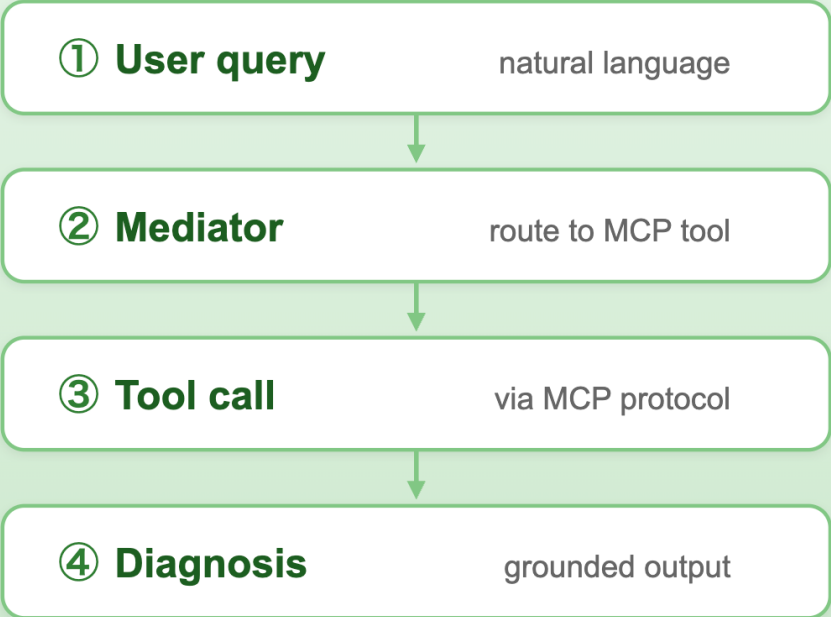
Same loop as research: define, equip, reason, act, check, repeat.

Path 2, Anatomy of the agent you are about to see



LangGraph Agent

A small LangGraph workflow



One X-ray, one question, a grounded radiology assessment.



SHAMSUL: Systematic Holistic Analysis to investigate Medical Significance Utilizing Local interpretability methods in deep learning

The SHAMSUL Demo App explores interpretability for chest X-ray pathology predictions using methods—Grad-CAM, LIME, SHAP, and LRP. It provides heatmaps and evaluation metrics for better insights into the medical significance of predictions made by deep learning models. For more details, see below.

Generated Images for All Original Classes

Ground...	Interpr...	Interpr...	Interpr...	Interpr...
Ground...	Interpr...	Interpr...	Interpr...	Interpr...
Ground...	Interpr...	Interpr...	Interpr...	Interpr...
Ground...	Interpr...	Interpr...	Interpr...	Interpr...
Ground...	Interpr...	Interpr...	Interpr...	Interpr...

Original Classes

Annotated Class Labels

- Support Devices
- Airspace Opacity
- Cardiomegaly
- Edema
- Atelectasis
- Enlarged Cardiomediastinum

Predictions

Predicted Class	Confidence	Decision
Support Devices	0.84	Correct
Airspace Opacity	0.52	Correct

Reflection, From Paper to Protocol



A paper describes a method. An agent runs one. The LLM is what makes that leap.

- A loop, not a script: the model chose the tools, the order, the arguments.
- "Just connecting APIs?" The calls are the easy half. A script encodes the path; the agent finds it.
- The LLM plans, interprets, recovers. That judgment is the agent; tools are its hands.
- The payoff: a goal in plain words, and the agent assembles the steps across your domains and data.
- The limit: it can be confidently wrong. Ground it, verify. Co-pilot, not oracle.

Agent-ready research infrastructure: serve.scilifelab.se/apps



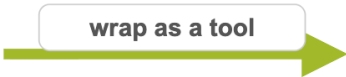
FROM ONE TOOL TO A PLATFORM

SHAMSUL

Chest X-ray pathology

+ Grad-CAM · LIME · SHAP · LRP

the one tool you just saw



156+

public research apps on SciLifeLab Serve
serve.scilifelab.se/apps

WHAT'S ALREADY DEPLOYED

Drug Discovery	Proteomics	Genomics & Transcriptomics	Spatial Biology & Imaging
Metabolomics & Exposomics	Clinical & Public Health demo	Ecology & Environment	Plant Biology
Metagenomics	Knowledge Graphs	Education	Data & General Tools

Connect and call the tool, not download and hope it runs.

Each one a candidate to become agent-ready.

Path 2, The builder's toolbox



You have choices. Today we pick one stack and build end to end.

- **Frameworks:** LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen. **Today: LangGraph (Session 1).**
- **The knowledge base:** any reasoning LLM, commercial or open. **Today: the SciLifeLab OpenLLM pilot. (Session 1 & 2)**
- **Sharing + deployment:** MCP to expose tools, Docker to package, SciLifeLab Serve to host. **Today: MCP (Session 2).**

Thank you! 😊



Next up: **Hands-on Session 1**

build your own agent with LangGraph and ReAct.

Contact:

datacentre@scilifelab.se, serve@scilifelab.se

The future belongs to those who adapt! 😊